Math 300 – Introduction to Mathematical Reasoning

Rutgers University, Summer 2023

Instructor:	Bernardo Rivas	**Office Hours:**	TTh 5pm–6pm, TIL–103A
Email:	bad162@math.rutgers.edu		W 5pm–6pm, Online via Zoom
Class meetings:	TTh 6pm–8pm, TIL-103A		

Canvas Course page: https://rutgers.instructure.com/courses/231430

Textbook:
We will be using the 3rd edition of the *Book of Proof* by Richard Hammack, 2018. ISBN: 978-0-9894721-2-8. This book is open access, so it can be downloaded as a PDF for free. A PDF is available on the Canvas page for this course, but you can also purchase a hard copy of the book for around $30.

Prerequisites:
Linear Algebra (640:250) or Multivariable Calculus (640:251) or equivalent, or permission from the department.

Course Description:
Math 300 (Intro to Math Reasoning) is a foundational course offered by the Math Department. This course is designed to introduce students to the fundamental concepts and techniques of abstract mathematical thinking and proofs, which serve as the backbone of advanced mathematics.

Throughout the course, students will develop their abilities to think critically, construct rigorous mathematical arguments, and communicate mathematical ideas effectively. Unlike previous courses, emphasis will be placed on logical reasoning, problem-solving strategies, and the development and analysis of mathematical proofs.

Course Warning: (or "cautionary message" available at Math 300 website)
Math 300 teaches students the thinking and communication skills of a professional mathematician. It is intended to prepare math majors and other students for Math 311 and Math 351, and more generally, to help them make the transition from the computational courses of the first two years to abstract courses based on reading and writing mathematical proofs.

Despite its number, Math 300 is not the easiest upper-level math course. Most students find it a tough course. In recent years, about a quarter of registered students either withdrew with a W or received a D or F in the course, but most of those students should never have registered. If you are not planning on taking any proof-based mathematics course you probably don't want to take Math 300. If this is the last math course that you plan to take, you should not take Math 300.

For the full cautionary message, please refer to the Math 300 website.

Academic Integrity:
All students in the course are expected to be familiar with and abide by the academic integrity policy. Violations of the policy are taken very seriously. See https://academicintegrity.rutgers.edu/ for more information.

Accommodations:
Full disability policies and procedures are indicated at http://ods.rutgers.edu/. Students with disabilities requesting accommodations must present a Letter of Accommodations to the instructor as early in the term as possible. See https://ods.rutgers.edu/my-accommodations/letter-of-accommodations.

Learning goals:

The learning goals for this class, as per the course website are:

> *Fundamental abstract concepts common to all branches of mathematics. Special emphasis placed on ability to understand and construct rigorous proofs.*

In particular, this consists of the following:

1. Students will understand what a mathematical proof is, identify logical flaws in incorrect proofs, and write well-structured proofs.
2. Students will learn propositional logic and how it applies to proofs and proof techniques.
3. Students will understand and be able to use a variety of proof structures and techniques, including, but not limited to
   a) Direct Proof
   b) Proof by Contradiction
   c) Proving the Contrapositive
   d) Proofs with sets and quantifiers
   e) Proof by Induction
   f) Pigeonhole Principle
4. Students will be able to compare and contrast the above proof methods and know how to choose which one should be used in a given case.
5. Students will be able to translate between English sentences and mathematical notation and be able to stitch between the two together in order to write and comprehend proofs.
6. Students will understand how examples fit into the process of proof creation and find counterexamples to mathematical statements.
7. Students will learn to effectively read and understand mathematics from a text.
8. Students will learn to typeset mathematical writing using LaTex.
9. Students will be prepared to go on to Math 311 and Math 351 and understand the proof in those classes.

Grading breakdown:

Grades will be calculated on the following scale:

Category	%
Attendance and Participation	5%
Homework	15%
Quizzes	20%
Midterm 1	15%
Midterm 2	15%
Final Exam	30%
Total	**100%**

Letter grades will be determined based on overall performance in the course. The cutoffs for each letter grade will not be determined until the end of the semester. Note that there is no 'curve' for this course, in the sense that there are no pre-set quotas for each letter grade. Thus, one student doing well does not disadvantage another student.

Note that you need a C or higher in this course to advance to 311, 350, or 351.

Attendance and Participation Policy:

Students are expected to attend all classes and be on time; if you expect to miss one or two classes or a period of time in a class, please use the University absence reporting website – https://sims.rutgers.edu/ssra/ - to indicate the date and reason for your absence. The system will automatically send an email to the instructor.

If for any reason a student will not be able to take an exam, or finds themselves in a situation, medical or otherwise, in which they will not be able to perform at their usual proficiency, they should notify the instructor right away and explain the situation. The instructor must be notified as soon as possible, and in any event before the exam. In the event of an extended absence, students should consult with the Dean of Students office to help verify their absence. Due to the accelerated pace of Summer courses, it is extremely difficult to catch up after falling behind.

To get full credit for a class meeting the student must be present throughout the meeting and actively participate in class activities. Each student can miss up to 2 class meetings without affecting their final participation grade.

Homework:

The homework will consist of weekly problem sets of varying difficulty that must be submitted through the course Canvas page. Students are encouraged and expected to collaborate on the problem set but will be required to write-up their solutions individually and in their own words. Each problem set must be typeset using LaTex. Writing in LaTex is an important skill in your mathematical toolbox and it is better to rip the bandaid off sooner than later. Late submissions or handwritten solutions will not be graded.

Quizzes:

There will be quizzes approximately once per week. Naturally, the quizzes will cover the specific topics or skills of the most recently submitted homework. Quizzes will be graded on a mastery-based system. That is, your quiz will score either one of four possible grades:

S Superior Mastery – The student demonstrates a comprehensive understanding of the topic, solves complex problems accurately, and constructs well-structured and logically sound statements.

A Advanced Mastery – The student shows a solid understanding of the topic, solves most problems correctly, and constructs coherent statements with minor errors.

B Building Proficiency – The student has a basic understanding of the topic, but struggles with more complex problems, makes significant errors, or lacks clarity in their statements.

C Cursory Proficiency – The student did not attempt the problem, or there isn't enough to make an accurate assessment of their understanding of the topic.

Each quiz grade will be assigned to one category or subcategory:

- Set theory
- Logic
- Proofs:
  - Direct proofs and counterexamples
  - Proofs by contrapositive and contradiction
  - Pigeonhole Principle
  - Proofs involving non-conditional statements
  - Proof by Induction
- Relations
- Functions
- Cardinality

You will be allowed **one reattempt** at each learning objective throughout the summer. In order to do this, you will need to:

1. Email me to let me know that you want to reattempt an objective, which objective you want to reattempt, and set up a time to meet with me during office hours.
2. Write up a completely correct solution, either typed or handwritten, to the quiz for the objective that you want to reattempt.
3. Meet me at the prescribed time to discuss the completed solution before you attempt the new quiz.
4. If the explanation is satisfactory, you may attempt a new quiz.

The same process holds if you have to miss a quiz for whatever reason. Each quiz can contribute up to 2% towards your final grade, for a total of 20% across all 10 quizzes.

- Each **S** adds 2% towards your total percentage.
- Each **A** adds 1% towards your total percentage.
- Each **B** adds 0.5% towards your total percentage.

Important Dates

Midterm 1 .... June 29
Midterm 2 .... July 27
Final Exam .... August 15

Exams:

There will be two Midterm Exams of 80 minutes and a Final Exam of 3 hours. The exams are cumulative, as everything in this course builds on itself. The main point of these exams is to emphasize the other components of the proof process: choosing which proof technique applies to a given problem, and carefully defend a mathematical argument. While quizzes cover a specific topic or skills, exams will cover a wider variety. The exams will be graded in a standard format, with partial credit and normal grading schemes.

Tentative Outline of the course:

Week	Date	Sections	Main Topics	Exam/Quiz
1	Tue, May 30	1.1-1.3, 1.5-1.6	Course introduction, LaTeX and Sets	Survey
1	Thu, Jun 01	1.4, 1.7, 1.9	Sets	-
2	Tue, Jun 06	2.1-2.6	Logic	Quiz (sets)
2	Thu, Jun 08	2.7-2.11	Logic	-
3	Tue, Jun 13	4.1-4.3, 9.1	Direct Proof and Counterexamples	Quiz (logic)
3	Thu, Jun 15	4.4-4.5	Direct Proof and Counterexamples	-
4	Tue, Jun 20	5.1, 6.1-6.2	Contrapositive/Contradiction	Quiz (proofs 1)
4	Thu, Jun 22	5.2, 6.3, 3.9	Contrapositive/Contradiction + Pigeonhole Principle	-
5	Tue, Jun 27	3.9	Review for midterm	Quiz (proofs 2)
5	Thu, Jun 29		N/A	Midterm + Quiz (proofs 3)
6	Tue, Jul 04		N/A	
6	Thu, Jul 06	7.1-7.2, 8.1-8.3	Proofs involving non-conditional statements & sets	-
7	Tue, Jul 11	7.3-7.4, 9.2-9.3	Proofs involving non-conditional statements & sets	-
7	Thu, Jul 13	10.1	Proof by induction	Quiz (proofs 4)
8	Tue, Jul 18	10.2-10.5	Proof by induction	-
8	Thu, Jul 20	11.1-11.5	Relations	Quiz (proofs 5)
9	Tue, Jul 25	11.6, 12.1-12.2	Relations and Functions	-
9	Thu, Jul 27		N/A	Midterm + Quiz (relations)
10	Tue, Aug 01	12.4-6	Functions	-
10	Thu, Aug 03	14.1-14.2	Cardinality	Quiz (functions)
11	Tue, Aug 08	14.3-14.4	Cardinality	
11	Thu, Aug 10	-	Review	Quiz (cardinality)
12	Tue, Aug 15		N/A	Final Exam

Disclaimer: This syllabus is subject to change according to the needs of the class, as deemed appropriate by the instructor. In case of changes, students will be notified in class and a new syllabus will be distributed.

Student Wellness Services

Counseling, ADAP & Psychiatric Services (CAPS)
(848) 932-7884 – 17 Senior Street, New Brunswick, NJ 08901
http://health.rutgers.edu/medical-counseling-services/counseling/
CAPS is a University mental health support service that includes counseling, alcohol and other drug assistance, and psychiatric services staffed by a team of professional within Rutgers Health services to support students' efforts to succeed at Rutgers University. CAPS offers a variety of services that include: individual therapy, group therapy and workshops, crisis intervention, referral to specialists in the community and consultation and collaboration with campus partners.

Violence Prevention & Victim Assistance (VPVA)
(848) 932-1181 – 3 Bartlett Street, New Brunswick, NJ 08901 – www.vpva.rutgers.edu/
The Office for Violence Prevention and Victim Assistance provides confidential crisis intervention, counseling, and advocacy for victims of sexual and relationship violence and stalking to students, staff and faculty. To reach staff during office hours when the university is open or to reach an advocate after hours, call 848-932-1181.

Disability Services
(848) 445-6800 – Lucy Stone Hall, Suite A145, Livingston Campus, 54 Joyce Kilmer Avenue, Piscataway, NJ 08854 – https://ods.rutgers.edu/
Rutgers University welcomes students with disabilities into all of the University's educational programs. In order to receive consideration for reasonable accommodations, a student with a disability must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation: https://ods.rutgers.edu/students/documentation-guidelines. If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with a Letter of Accommodations. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. To begin this process, please complete the Registration form on the ODS web site at: https://webapps.rutgers.edu/student-ods/Forms/LOA.

Report a Concern: http://health.rutgers.edu/do-something-to-help/